

waveforms, including single-pulse and pulse-compression capabilities, for radar sensing. It integrates two RISC-V microprocessors, providing perfor-



mance and flexibility for applications. With 3D beamforming, hydrogen can control up to four transmit and four receive antennas, allowing for array configurations. Its architecture ensures synchronisation across channels, enabling pulse focusing in space—something 60GHz devices struggle with. Hydrogen allows 3D localisation with depth perception, making it suitable for gesture recognition, ceiling-mounted presence sensors, and in-cabin vehicle detection.

ARIA Sensing

<https://ariasensing.com>

AI at the edge with NPU-accelerated microcontrollers

STMicroelectronics has introduced the STM32N6 microcontroller (MCU) series, its first to integrate a neural processing unit for machine learning. This enables consumer and industrial



products to support features like computer vision, audio processing, and sound analy-

sis, which were previously beyond the reach of embedded systems. The company claims that it is the most capable MCU to date, featuring the Neural-ART accelerator, which delivers 600 times the ML performance of a high-end STM32 MCU. The microcontroller series can be used by industrial IoT engineers and embedded systems designers for applications like computer vision, audio processing, and sound analysis. AI and machine learning developers can use its compatibility with frameworks, while manufacturers of vision-related products can use its integrated image signal processor for image sensor processing.

STMicroelectronics

<https://www.st.com>

Industry's first multi-modal GenAI edge product

SiMa.ai has launched the Modalix 50 TOPs device. It delivers over 10 times the performance per watt compared



to alternatives. The MLSoC architecture enables real-time human-machine interactions while processing multiple sensor inputs, making it suitable for applications requiring complex decision-making. Modalix builds on SiMa.ai's first-generation MLSoC, supporting AI and GenAI use cases on-device at under 10 watts. It brings GenAI and conversational AI to various industries, including robotics, automotive infotainment, smart factories, retail, medical, aerospace, defence, agriculture, construction, and industrial drones.

SiMa.ai

<https://sima.ai>

Industry's first PCIe 6 switch

Astera Labs has introduced the Scorpio Smart Fabric Switch portfolio, featuring the industry's first PCIe 6 switch designed specifically for AI



workloads in cloud-scale computing. These fabric switches optimise AI dataflows for efficient performance, high reliability, and simplified cloud deployment while reducing costs and time-to-market. The smart fabric switches maximise system utilisation for AI workloads, offering energy-efficient, high-performance fabric solutions with advanced fleet management. The company claims that it is purpose-built for AI, addressing these challenges with a more efficient and scalable fabric solution.

Astera Labs

<https://www.asteralabs.com>

Industry's first PCIe Wi-Fi 7 + Bluetooth combo module

Silex Technology has launched the SX-PCEBE, the first embedded radio module supporting Wi-Fi Certified 7 (802.11be) and Bluetooth 5.3. The



module delivers dual-band simultaneous operation across 2.4GHz, 5GHz, and 6GHz for connectivity. The company claims that the module is designed for AI-powered devices in medical and industrial applications, providing plug-and-play Wi-Fi 7 connectivity for platforms from Nvidia, Qualcomm, and NXP. The SX-PCEBE uses Wi-Fi 7's wider channels and higher capacity, delivering twice the throughput of Wi-Fi 6, improved reliability with multi-link operation, and 20% faster transmission with reduced latency.

Silex Technology

<https://www.silextechnology.com>

New gas-leak sensor

Interlink Electronics has introduced a new addition to its electrochemical gas sensor portfolio. The 110-37x sensor family is designed to detect TBM (tert-



butyl mercaptan) and THT (tetrahydrothiophene), compounds added to natural gas pipelines to give odourless methane a detectable smell for safety purposes. These sensors provide an automated way to detect gas leaks in residential, commercial, and industrial environments. The company states that the sensors use electrochemistry while maintaining performance, small size, and cost efficiency. Although these gases can be detected by smell, an automated detection solution is necessary for improving safety in natural gas applications.

Interlink Electronics

<https://www.interlinkelectronics.com>